

QFlow Nexys3 Fast Operation and Recovery Runbook v2

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Board: Digilent Nexys3 Rev. B, Spartan-6 XC6SLX16-CSG324

Purpose: Fast repeatable board startup, raw-FX2 recovery, and safe QFlow programming

Updated: 19 July 2026

1. Executive rule

Do not repeat the full EEPROM investigation during normal operation. The official 4,760-byte Digilent controller image has already been verified by multiple exact readbacks. A block of 0xCD is VendAX transfer failure-fill, not erased EEPROM.

Routine recovery stops as soon as the board returns to operational USB identity 1443:0007. Never write the controller EEPROM unless repeatable stored-byte corruption is independently proven and explicitly reviewed.

2. Verified board identity

Item	Verified value
Operational USB	1443:0007
Raw Cypress recovery USB	04b4:8613
Adept product	Nexys3
Adept serial	210182697240
FPGA	XC6SLX16
JTAG ID	44002093
Repository branch	rl-nexys3-adaptive
Baseline HEAD before final repair commit	2b6f8fdd7465e0b2135a3db8ae57cdc2ed1d431a

Rev. B direct-I/O mapping verified from board silkscreen

Input	FPGA pin	D diagnostic LED	LED pin
SW0	T10	LD5	N11
CENTER / BTNS	B8	LD0	U16
UP / BTNU	A8	LD1	V16
LEFT / BTNL	C4	LD2	U15
DOWN / BTND	C9	LD3	V15
RIGHT / BTNR	D9	LD4	M11

LD6/R11 is the loaded-design marker and LD7/T11 is the heartbeat in the direct diagnostic. All six inputs passed when internal FPGA PULLDOWN constraints were removed.

3. Normal power-on workflow

1. Ensure the ISE image is mounted read-only at /mnt/xilinx-ise14.7.
2. Connect the Nexys3 USB cable and turn the board power ON.
3. Wait 5 seconds.
4. Run clean USB/Adept enumeration with all ISE library variables removed.
5. If 1443:0007, program the checksum-verified .bit file using system /usr/bin/djtgcfg.
6. Keep the board powered for the entire FPGA test session because FPGA SRAM is volatile.

Clean enumeration

```
env -u LD_LIBRARY_PATH -u LIBRARY_PATH \  
-u XILINX -u XILINX_DSP -u XILINX_EDK \  
/usr/bin/lsusb -d 1443:0007
```

```
env -u LD_LIBRARY_PATH -u LIBRARY_PATH \  
-u XILINX -u XILINX_DSP -u XILINX_EDK \  
/usr/bin/djtgcfg enum
```

4. Fast raw-mode recovery

A. One deep cold restart

Use this when 04b4:8613 appears:

1. Turn board power OFF.
2. Disconnect USB and every external power source.
3. With all sources disconnected, move the power switch ON for 30 seconds to discharge the board, then return it OFF.
4. Reconnect USB.
5. Turn board power ON and wait 10 seconds.
6. Check for 1443:0007.

If operational identity returns, stop recovery and program the FPGA. Do not load VendAX and do not read EEPROM.

B. If raw mode persists

Use the verified VendAX helper only in volatile FX2 RAM:

- VendAX SHA-256: 0e4e5da410b51eb7a111cb8d3627fac02eada997916fefc9ea56f4fc9a673dd9
- Expected load: 3,359 bytes in 41 segments.
- Run A3 RAM self-test at 0x064D, length 16.
- Send AA twice to select 100 kHz.
- Make one 16-byte A9 prefix probe.
- Expected prefix: c2 43 14 0d 00 00 00 00 00 04 00 00 02 00 6b 32.

If the prefix is exact, the EEPROM path is readable. Stop diagnostics and perform another deep cold restart. Full two-pass EEPROM reads are incident diagnostics, not a normal startup requirement.

C. Stop condition

If the stored image is known exact but raw mode repeatedly returns, classify the problem as intermittent FX2 boot/power-sequencing reliability. Do not keep rewriting or repeatedly reading EEPROM. Use the next operational window, keep the board powered, and finish all FPGA tests in that session. For a long-term hardware solution, use an external JTAG cable or another Nexys3 board.

5. Evidence-based decision table

Observation	Meaning	Action
1443:0007, Adept finds Nexys3	Controller operational	Program verified bitstream
04b4:8613 after normal start	FX2 raw boot	One deep cold restart
A3 mismatch	VendAX RAM helper invalid	Stop; do not access EEPROM
A3 exact, A9 prefix exact	Helper and EEPROM read path good	Restart; no EEPROM write
A9 returns all 0xCD	Transfer failure-fill	Wait, reprobe once; do not infer erasure
One full pass exact, another differs	Intermittent read transport	Accept exact stored-image evidence; stop repeated reads
Repeated stored bytes differ at stable offsets	Possible corruption	Preserve evidence and require review before any write

6. Current P5 B1 repair state

The original B1 UCF incorrectly enabled internal PULLDOWN on the externally driven center button. The no-pulldown C3C diagnostic proved:

- SW0 controls LD5.
- CENTER, UP, LEFT, DOWN, and RIGHT control LD0 through LD4.
- LD6 is steady ON and LD7 blinks.
- All input LEDs are OFF in low/released state.

The repaired real B1 build removed only the active PULLDOWN constraint from `btn_reset` at B8. It passed NGDBuild, Map, PAR, Trace, and BitGen with zero timing errors.

Repaired B1 artifact	Verified value
Bitstream bytes	464,294
Bitstream SHA-256	See exact hash below
Minimum period	6.253 ns
Maximum frequency	159.923 MHz
Timing score	0

7eefc2e1cae4089cc47e58c2cca11a8b91efa359630d2e3b70fefe1074bc5347

The repaired bitstream must still receive one final physical reset validation: display counts, CENTER holds display at 0000, LD1 indicates reset, and counting restarts from 0001 after release.

7. GitHub preservation plan

Commit and push these reusable, non-proprietary assets after the physical gate:

- This runbook in Markdown and PDF.
- Board identity and pin map.
- Normal enumeration and programming scripts.
- Raw-mode detection and safe recovery wrapper.
- A3/AA/16-byte probe script.
- Repaired B1 UCF and deterministic bitstream.
- Sanitized evidence manifests and SHA-256 values.
- A note explaining that 0xCD is transfer failure-fill.

Do not commit the Xilinx license, the 50 GiB ISE image, USB serial logs containing unnecessary host details, or proprietary Digilent/Cypress firmware bytes to a public repository. Store paths, hashes, retrieval instructions, and locally backed up recovery binaries instead.

8. Current incident conclusion

On 19 July 2026, a fresh VendAX session produced an exact A3 result and an exact first 4,760-byte EEPROM pass with SHA-256 `ad744e1908ab029b9c08400dbab49f00998c0da0300fa16f188782566893c6b8`. A later pass differed, confirming intermittent transport rather than loss of the verified official image. No EEPROM write was performed.

The operational priority is therefore to obtain one clean controller boot, keep the board powered, program the already-built repaired B1 bitstream, finish its physical reset validation, preserve the reusable assets, and close P5 Step 3.